



Department of Master of Computer Applications

DESIGN THINKING LAB – MCA427DL

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TITLE: Efficient Beekeeping & Hive Box — Safe Inspection & Feeding Upgrade

EMPATHY → DEFINE



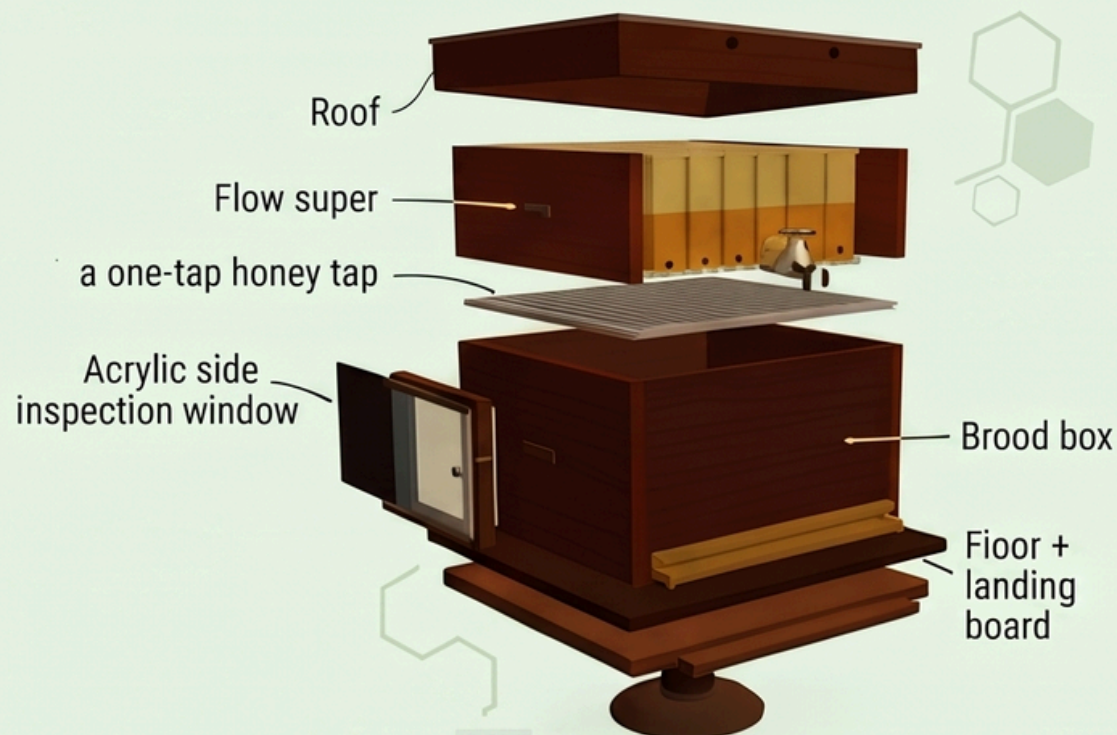
- 7 interviews · field visits · survey + secondary research
- 19 pain points mapped
- beginners fear stings; frames glued by burr-comb

“Opening the box ruins the hive’s temperature for two days.”

small-scale & novice beekeepers need a **safer, simpler way to inspect & maintain hives.**



PROTOTYPE



INLAYS



Acrylic Inspection window (UV cover) [based on viIMAGE 1+]



Assembled on elevated stand [IMAGE 6]



Closed-loop top feeder (funnel) [IMAGE 7]

Key specs: <₹500 retrofit · ISI standard · 9 mm bee space · inspect & feed without opening

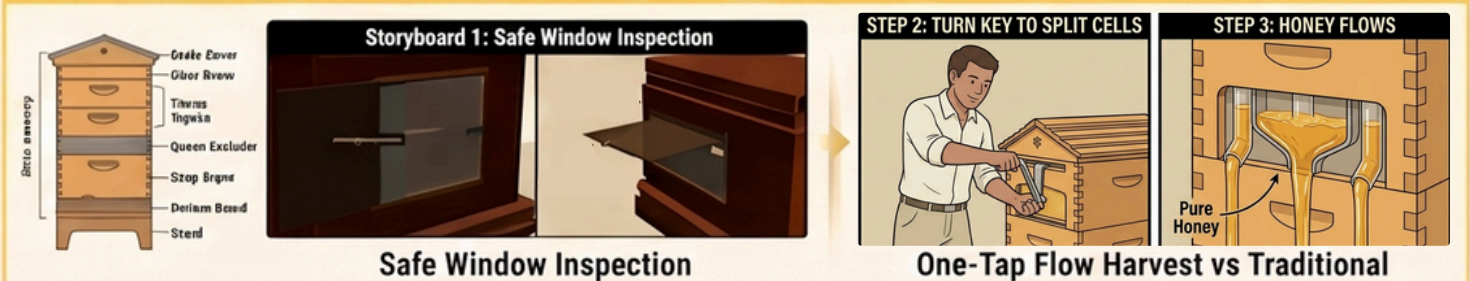
JOURNEY RIBBON



IDEATE

- bullets — acrylic inspection window (UV sliding cover), external gravity sugar-syrup feeder, · one-tap flow auto-harvest · donut ant-barrier / water ring

★ Selected: Safe Inspection & Feeding Upgrade — low-cost (<₹500) retrofit



Detailed Safe Feeding Process in Prototype section.



Safely Feeding without opening · Detailed Process

TEST & IMPACT

Concept validation (pre-field stage)

Method

- 3D model validation
- dimensional check vs IS 1515:1998 standard
- expert review with interviewed beekeepers

Check

- ✓ inspect?
- ✓ feed?
- ✓ retrofits?
- ✓ <₹500?
- ✓ ISI standard?

Outcome

- validated dimensions
- expert feedback on simplified
- requirement checklist met
- Planned 6-week field test

Beneficiaries

- small/novice beekeepers
- elderly & women farmers
- FPOs

SDGs Addressed



Expected Outcome

- cuts colony loss & raises yield
- 20-min - 30-sec safe check
- accessible to all skill levels

REFERENCES IS 1515:1998 (BIS); Monga & Manocha (2011), LRRD 23(5); Gadekar (2026), I.IEMR 16(1); Flow Hive (Wikipedia); MANAGE – Scientific Beekeeping for Livelihood Security; Agriculture.Institute (2025)

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